

# Algorithmic Accountability Guideline

PURPOSE

Public declaration of model logic, API dependencies, and data sovereignty. ARISE Framework — Pillars 4 & 6. Procurement Ready.

## 1. System Description

| Field               | Value                                                                                    |
|---------------------|------------------------------------------------------------------------------------------|
| System Name         | Dëkkal Flood Risk Scoring Engine v1.0                                                    |
| Type                | XGBoost Gradient Boosting Classifier (Statistical Layer)                                 |
| Purpose             | Deliver address-level flood risk scores (0-100) to IARD insurers in Dakar, Senegal       |
| Output              | Risk score + risk level + 3 explanatory factors + confidence interval + decision support |
| Coverage            | Dakar urban zone                                                                         |
| Responsible Officer | Babacar Ndao, CTO                                                                        |

## 2. Model Logic

The scoring engine uses a three-component weighted aggregation:

| Component                | Weight | Description                                                                                    |
|--------------------------|--------|------------------------------------------------------------------------------------------------|
| Historical Risk          | 25%    | Sentinel-1 SAR water detection delta — satellite-confirmed flood signals                       |
| Structural Vulnerability | 35%    | SRTM elevation, slope, clay fraction (SoilGrids), ERA5 soil moisture, OSM drainage proximity   |
| Extreme Scenario Risk    | 40%    | CHIRPS + GPM P99 rainfall vs drainage capacity proxy (25mm/day) — forward-looking climate risk |

## 3. Third-Party API Dependencies

| Service             | Provider           | Purpose                                                     | Data Sovereignty                      |
|---------------------|--------------------|-------------------------------------------------------------|---------------------------------------|
| Google Earth Engine | Google LLC         | Feature extraction (SAR, DEM, CHIRPS, ERA5, SoilGrids, GPM) | Processed on GEE servers — not stored |
| Sentinel-1 SAR      | ESA / Copernicus   | Flood detection imagery                                     | Open — CC BY SA                       |
| CHIRPS Daily        | UCSB / USAID       | Precipitation time series                                   | Open license                          |
| ERA5-Land           | ECMWF / Copernicus | Soil moisture                                               | Copernicus licence                    |

| Service              | Provider         | Purpose                   | Data Sovereignty                |
|----------------------|------------------|---------------------------|---------------------------------|
| SoilGrids            | ISRIC            | Soil clay/sand properties | CC BY 4.0                       |
| GPM IMERG            | NASA EarthData   | Extreme rainfall P99      | Open license                    |
| Nominatim            | OpenStreetMap    | Address geocoding         | ODbL                            |
| Sénégal Numérique SA | Senegalese State | Data hosting              | ISO 27001 certified — sovereign |

4. Performance Metrics (LOO-CV on 40-sample dataset)

| Metric           | Value | Target              | Status      |
|------------------|-------|---------------------|-------------|
| Accuracy         | 80.0% | —                   | Baseline    |
| Precision        | 89.3% | >75%                | ✓ Achieved  |
| Recall           | 83.3% | >80%                | ✓ Achieved  |
| F1-Score         | 86.2% | >77%                | ✓ Achieved  |
| Training samples | 40    | Target: 200+ (v1.1) | In progress |

5. Known Limitations

- Training dataset: 40 samples — XGBoost performance will improve with more samples
- SRTM 30m resolution: limited address-level discrimination in flat urban zones
- LSTM temporal layer: in development for v2.0 — not deployed in v1.0
- Coverage: Dakar urban zone only — no other Senegalese cities yet

6. Data Sovereignty Statement

All Dëkkal data assets are hosted exclusively at Sénégal Numérique SA, Diamniadio, Senegal — ISO 27001 certified. No personal data or client query data is transferred outside West Africa.